AnchorMap: A Multi-Agent Pipeline for Variable Standardisation in Maritime Engineering

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ABSTRACT

Maritime co-simulation and collaborative design require different organisations to exchange model data, and equivalent physical quantities are often labeled differently across Original Equipment Manufacturer (OEM) datasheets, simulation models, and regulatory specifications because each organization conceptualizes its domain independently. This causes variable-name mismatches that hinder co-simulation. This paper presents AnchorMap, a schema-driven multi-agent pipeline for standardising heterogeneous variable labels to a canonical vocabulary. A JSON for Linked Data (JSON-LD) schema aligned with the Vessel Information Structure (VIS) defines canonical variable identifiers with metadata and provenance. AnchorMap combines embedding-based candidate retrieval with conditional large language model (LLM) reasoning and applies a confidence-based routing policy to accept reliable mappings, abstain when no justified match exists, or defer ambiguous cases to human review. The approach is evaluated on 144 variable labels from electric motor catalogues and research publications. At the selected operating point, the system achieves a precision of 0.787, recall of 0.842, and F1-score of 0.814, while routing uncertain cases to human review. Results demonstrate that confidence-aware routing enables controlled automation of variable standardisation, and highlight that LLM-based reasoning may select plausible candidates even when no justified match exists, which motivates improved abstention and calibration mechanisms in future work.

INTRODUCTION

Collaborative ship design requires shipyards, OEMs, class societies, and regulators to exchange models and specifications, but equivalent quantities are labeled inconsistently across sources (e.g., "MCR Power", "P_ME", "Max Engine Power (kW)"). These mismatches obstruct standards-based workflows such as Functional Mock-up Interface (FMI) and System Structure and Parameterisation (SSP), where reliable co-simulation and compliance checking depend on consistent parameter identifiers. VIS provides a shared vessel vocabulary, yet a practical gap remains at the variable level: the same concept appears under different abbreviations, unit-bearing labels, and subsystem-dependent conventions.

Previous work has improved semantic interoperability at higher levels, for example by adding ontology-based semantic types for FMI variables in maritime co-simulation (Rindarøy et al., 2025), and by using LLMs for schema/ontology alignment when combined with embedding retrieval and uncertainty handling (He et al., 2023). However, engineering variable names in OEM documentation remain challenging to standardize automatically because labels are short, context specific, and often ambiguous, increasing the risk of confident but unjustified matches.

This paper presents AnchorMap, a schema-driven multi-agent pipeline that maps heterogeneous OEM variable labels to canonical identifiers defined in a VIS-aligned JSON-LD schema. The system combines embedding-based candidate retrieval with conditional LLM reasoning and routes outcomes by confidence to *Accept*, *Human Review* or *No_Match*. We evaluate the approach on 144 variable labels from electric motor catalogues and literature sources and analyse the automation–review trade-off under different routing thresholds. The main contributions of this work are:

1. Develop an automated mapping pipeline that aligns heterogeneous OEM variable labels to canonical identifiers using embedding-based candidate retrieval and LLM-based reasoning.

2. Construct a structured JSON-LD schema for ship hull and propulsion systems that provides a consistent set of canonical engineering variables aligned with the Vessel Information Structure (VIS).

3. Validate the proposed approach through an electric motor subsystem case study involving variable la-

bels from multiple OEM catalogues and specification sheets.

RELATED WORK

Standards for collaborative co-simulation such as FMI and SSP define how models and architectures are packaged, exchanged, and parameterised (Blochwitz et al., 2011; Hällqvist et al., 2021). Open Simulation Platform Information Model Specification (OSP-IS) extends FMI with an ontology that groups variables into domain-relevant semantic types (Rindarøy et al., 2025), while other efforts standardise how requirements are authored and shared in this environment (Al-Shami et al., 2025). Broader frameworks for automated multi-vendor mechanical system design similarly rely on standardised Digital Twin documents distributed through a Digital Twin Web (Ala-Laurinaho et al., 2025). Despite this infrastructure, a gap remains at the variable level: OEMs use different names, abbreviations, and unit conventions for the same physical quantities. This gap is difficult to close through standardisation alone because OEMs have limited incentive to change existing practices, and ontology/naming-schema adoption often requires substantial modelling effort and manual alignment (Spoladore and Pessot, 2022). AnchorMap addresses this adoption barrier by treating a JSON-LD schema as a fixed target vocabulary and automating alignment, while routing uncertain cases to human review to reduce deployment risk.

Schema and ontology alignment methods range from lexical matching to embedding and LLM-based reasoning. String and lexical similarity metrics handle surface variations but fail when equivalent entities share few lexical features, common in engineering labels (e.g., “P_ME” vs. “Max Engine Power”) (Cheatham and Hitzler, 2013). LLM-based approaches can reason over candidate alignments and have shown competitive performance in ontology matching (He et al., 2023); they can also be used as rerankers over candidate sets to balance accuracy and cost (Liu et al., 2024). More structured programs combine candidate generation, confidence scoring, and refinement to improve alignment (Seedat and van der Schaar, 2024), and applied retrieval-then-reasoning pipelines pair embedding retrieval with LLM accept/reject decisions in domain settings (Kokash et al., 2025). However, lexical and embedding methods can still produce false matches between similar-looking variables, and LLM reasoning may select plausible candidates when no justified target exists. This motivates a pipeline that combines retrieval and constrained reasoning with explicit routing of uncertain cases via dual confidence thresholds.

Table 1 compares AnchorMap qualitatively with existing schema and LLM-based alignment approaches across six aspects of the pipeline. The comparison highlights that AnchorMap achieves more reliable matching of abbreviated and context-dependent OEM labels, and that it can be integrated directly into an engineering workflow. AnchorMap also explicitly routes uncertain cases to *Human Review* or *No_Match*, which reduces the risk of incorrect variable renaming in the output Turtle files.

Table 1: Qualitative comparison of AnchorMap with representative alignment practices

Aspect	Lexical / Embedding	LLM-Based Alignment	AnchorMap
Handling of abbreviated or paraphrased labels	Struggles with short or dissimilar labels that refer to the same quantity	Can use contextual information, but performance depends on prompt design and candidate quality	Combines candidate retrieval with contextual reasoning to resolve abbreviated OEM labels
No justified match exists	Nearest candidate may still be returned	Plausible but unjustified candidate may still be selected	Explicit *No_Match* and Human Review routes via dual confidence thresholds, reducing the risk of false matches
Uncertainty handling	Usually implicit or based only on similarity score	Often prompt-dependent and difficult to interpret consistently	Uses two confidence thresholds to route each case to Accept, Human Review, or *No_Match*
Human-in-the-loop support	Usually absent	Sometimes added externally	Built in as an explicit human-review step before renaming is applied
Verification & Validation (V&V)	Limited, because incorrect matches may pass silently	Potentially useful, but risky without abstention controls	Designed for practical co-simulation and V&V workflows, where false accepts are costly
Use-case grounding	General-purpose matching	General-purpose alignment reasoning	Grounded in a VIS-aligned JSON-LD schema for maritime engineering variable standardisation

Prior alignment methods primarily optimise matching quality, whereas AnchorMap additionally addresses the decision boundary required for practical deployment: whether a candidate should be accepted automatically, rejected as *No_Match*, or escalated for human judgement. This distinction is important in the present setting because accepted mappings trigger identifier changes in engineering artefacts used in co-simulation and compliance-related workflows.

METHODOLOGY

Schema Design and JSON-LD Representation

A schema was defined to provide standard identifiers for engineering variables and to bind each variable to an explicit subsystem context. The schema follows the hierarchical system structure defined by the Vessel Information Structure (VIS), so that each quantity is associated with the ship object it describes such as prime mover, shafting, propeller, motor. VIS is specified in ISO 19848 (International Organization for Standardization, 2024) as a codebook that provides unique identification strings for Internet of Things (IoT) sensors and data sources on ships, and the schema in this work extends that identification principle to engineering variables used in simulation and design. Multiple stakeholders contribute simulation models and data in collaborative ship design, and variable naming remains consistent only if each participant resolves their labels against the same reference. The JSON-LD schema therefore serves as the project’s Single Source of Truth (SSOT) for standard variable identifiers.

Table 2: Canonicalisation Rules for Schema Variable Identifiers

Rule	Applied To	Canonicalisation Principle
R1	All variables	Convert to `snake_case`, remove punctuation, and apply minimal token reordering only when unambiguous.
R2	Rated values	Use the `rated_` prefix for nameplate quantities (e.g., `rated_current`).
R3	Unit-bearing quantities	Keep identifiers unit-free and store units as metadata (`unitSymbol` or a unit Internationalized Resource Identifier (IRI)).
R4	Context-sensitive terms	Disambiguate via domain placement (schema context), not by adding subsystem tags to the identifier.
R5	Performance metrics	Use explicit qualifiers (e.g., `_efficiency`, `_loss`) and encode operating conditions only when needed for uniqueness.
R6	Categorical properties	Model as categorical values (string/enum), not as numeric quantities with units.
R0	Out-of-scope	Exclude non-standard or vendor-specific terms and treat as *`No_Match`* during mapping.

The schema is encoded in JSON-LD, preserving Resource Description Framework (RDF) semantics through explicit contexts and identifiers (Kellogg et al., 2020). Engineering variables are represented as RDF properties with unique identifiers (`@id`), subsystem assignments via `rdfs:domain`, and expected datatypes via `rdfs:range`. Domain assignment separates variable meaning from subsystem context, allowing identifiers such as `rated_current` to be reused across components while remaining contextually disambiguated. Units and standard references are stored as metadata to preserve traceability without embedding unit conventions in the identifier.

Canonical identifiers were constructed using a rule-guided procedure: variables are normalised into a consistent naming form, ambiguity is resolved through schema context rather than by embedding subsystem tags into identifiers, and units are represented as metadata to keep identifiers stable across unit conventions. The canonicalisation rules applied in this work are summarised in Table 2.

The schema extends VIS with motor-specific terms grounded primarily in the International Electrotechnical Commission (IEC) 60034 series and related IEC and International Organization for Standardization (ISO) standards (International Electrotechnical Commission, 2022, 2014a, 2016, 2014b,c, 1991, 2007; International Organization for Standardization, 2009). Provenance is recorded via `dcterms:source` to support traceability.

The schema does not attempt to enumerate all OEM label variants. Out-of-scope or vendor-specific terms are routed to *No_Match* rather than forced into incorrect mappings. For retrieval, each schema term is represented as a compact anchor card containing identifier, definition, domain context, datatype, and metadata. The schema is versioned to ensure reproducibility.

AnchorMap: Multi-Agent Variable Standardisation Pipeline

AnchorMap standardises heterogeneous OEM variable labels to the canonical identifiers defined in the JSON-LD schema by combining embedding-based retrieval, conditional LLM reasoning, and confidence-based routing. The end-to-end flow is shown in Figure 1. The pipeline takes as input OEM files encoded in Turtle format containing OEM-defined variable labels and produces a Turtle output file in which accepted labels are replaced by schema-aligned canonical identifiers while preserving the original RDF graph structure and document metadata.

The Schema Indexing Agent prepares the reference space used for all subsequent matching. The JSON-LD schema is treated as a collection of canonical variable definitions, where each definition is treated as a self-contained schema element. The Schema Indexing Agent chunks the schema into schema elements and computes embeddings for each element using the OpenAI text-embedding-3-large model to generate dense vector representations. The resulting vectors form the anchor space against which OEM variables are compared. The Retrieval and Reasoning Agent processes each OEM variable independently. The pipeline reads OEM Turtle (RDF) files and extracts variable definitions using SPARQL Protocol and RDF Query Language (SPARQL). For each input label, the agent computes an embedding and retrieves Top-K schema candidates by cosine similarity, yielding a ranked set $\{(a_i, s_i)\}_{i=1}^{K}$ where a_i is a candidate identifier and $s_i \in [0, 1]$ is its similarity score. Here s_1 and s_2 denote the top and runner-up similarity scores, τ_{sim} is the similarity cutoff, τ_{gap} is the minimum separation margin, c is the scalar confidence used for routing. τ_{nm} and τ_{hr} are the *No_Match* and accept thresholds, respectively.

To avoid unnecessary LLM calls, the agent applies a deterministic shortcut: if the top candidate is sufficiently strong ($s_1 \geq \tau_{\text{sim}}$) and sufficiently separated from the runner-up ($(s_1 - s_2) \geq \tau_{\text{gap}}$), it selects a_1 directly and sets $c = s_1$. Otherwise, the agent invokes constrained LLM reasoning over the retrieved candidates and sets $c = c_{\text{llm}}$. We use GPT-4.1 for reasoning and `text-embedding-3-large` for retrieval.

Routing uses two thresholds on the scalar confidence c: $c \leq \tau_{\text{nm}} \Rightarrow$ *No_Match*, $\tau_{\text{nm}} < c \leq \tau_{\text{hr}} \Rightarrow$ Human Review, and $c > \tau_{\text{hr}} \Rightarrow$ Accept.

For Human Review, the reviewer is shown the input label, the retrieved candidates with scores, and the corre-

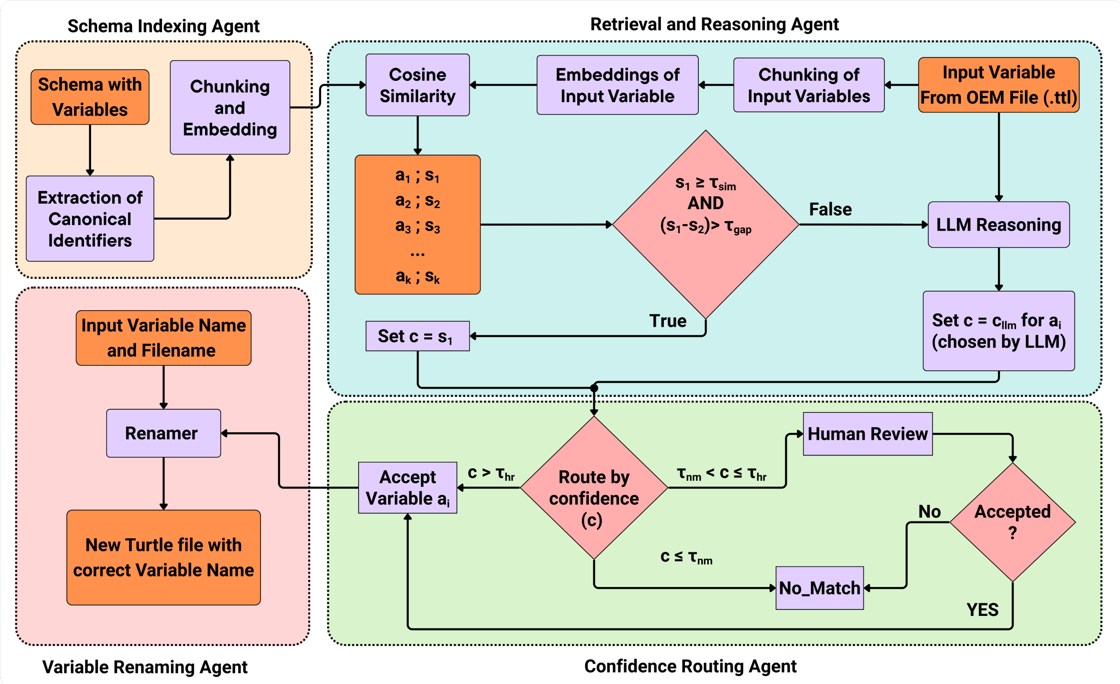

Figure 1: AnchorMap Multi-Agent Variable Standardisation Pipeline: Schema Indexing Agent builds the anchor space from the JSON-LD schema. Retrieval and Reasoning Agent retrieves candidates by cosine similarity and uses LLM reasoning when needed. Confidence Routing Agent applies dual thresholds to route Accept, Human Review, or *No_Match*. Variable Renaming Agent rewrites accepted labels in the OEM Turtle (.ttl) file to produce schema-aligned output.

sponding schema content, and either approves one candidate (Accept) or rejects all (*No_Match*).

The Variable Renaming Agent rewrites the original Turtle file by replacing OEM labels with the selected canonical identifiers for variables routed as Accept. Variables routed as *No_Match* are left unchanged, and Human Review cases are rewritten only after approval, preserving the original RDF structure and metadata.

EXPERIMENTS

This evaluation is conducted in a requirements-driven motor specification standardisation setting, where motor requirements are authored in Requirements Temporal Logic Ontology (ReqTLo) (Al-Shami et al., 2025). Candidate motor specifications are represented as Turtle artefacts collected from heterogeneous OEM and literature sources, and AnchorMap is used to standardise their variable labels to the canonical identifiers in the JSON-LD schema.

Experimental Design

The experiments evaluate AnchorMap as a selective automation pipeline for variable label standardisation. Each input label is routed to one of three outcomes: Accept (the label is rewritten to a canonical schema identifier), Human Review (the system defers the decision), or *No_Match* (the label is left unchanged). We report the accuracy of the subset of decisions made automatically and quantify the review workload induced by different confidence thresholds. A threshold sweep over τ_{hr} is used to characterise the trade-off between automation and human review. Concretely, a lower threshold ($\tau_{\text{hr}} = 0.45$) maximises recall (0.883) but accumulates many false accepts (35 FP), whereas a higher threshold ($\tau_{\text{hr}} = 0.55$) increases precision (0.815) but routes most cases to human review (71.5%). Routing follows the dual-threshold confidence policy illustrated in Figure 1.

Dataset

The evaluation dataset contains variable labels extracted from nine Turtle artefacts: five derived from manufacturer catalogues ((Nidec Leroy-Somer, 2023; Bonfiglioli S.p.A., n.d.; WEG S.A., 2020; Leroy-Somer, 2008; ABB Oy, 2004)) and four derived from literature sources ((Hinkkanen and Luomi, 2004; Aarniovuori et al., 2016; Laine et al., 2025; Szewczyk, 2024)). In total, $N = 144$ variable-label instances were evaluated. Ground-truth mappings were provided for all instances. Of these, 123 instances have a canonical schema identifier and should be mapped, whereas 21 instances have no canonical target in the schema and should be rejected as non-canonical or vendor-specific.

Ground truth was produced by manually mapping each instance to the most appropriate schema identifier when a canonical target existed; otherwise it was labelled as out-of-scope. The out-of-scope instances correspond to

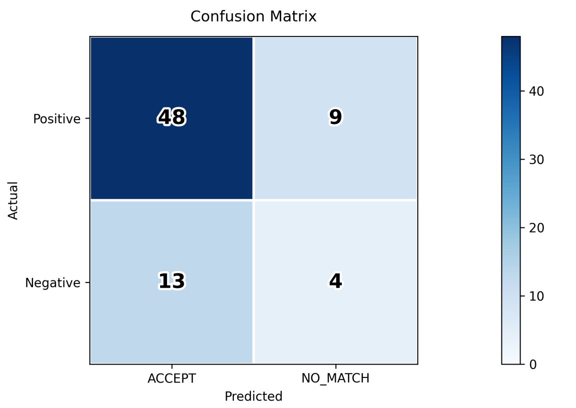

Figure 2: AUTO-Only Confusion Matrix at $\tau_{\mathrm{hr}} = 0.50$ (Accept vs. *No_Match*, Human-Review Items Excluded)

vendor-specific or non-standard quantities not represented in the schema.

Evaluation metrics

We evaluate performance on the subset of instances that receive an automatic outcome in $\{$*Accept*, *No_Match*$\}$. This automatic-decision-only (AUTO-only) evaluation excludes instances routed to *Human Review*. *Accept* is treated as a positive prediction and *No_Match* as a negative prediction. A true positive (TP) requires both routing to *Accept* and selecting the correct canonical identifier (exact match to ground truth). Precision, recall, and F1-score are computed from the resulting confusion matrix. Human review workload is reported as N_{HR} and as the human-review rate $\mathrm{HR}(\%) = 100 \cdot \frac{N_{\mathrm{HR}}}{N}$, where N is the total number of inputs.

Implementation details

We sweep $\tau_{\mathrm{hr}} \in \{0.45, 0.50, 0.55, 0.60, 0.65\}$ and hold all other parameters constant: $K = 5$, $\tau_{\mathrm{sim}} = 0.45$, $\tau_{\mathrm{gap}} = 0.06$, $\tau_{\mathrm{nm}} = 0.40$, and $T = 0.1$.

Main results

We report AUTO-only precision, recall, and F1 on the subset of instances that receive an automatic outcome in $\{$*Accept*, *No_Match*$\}$, excluding items routed to *Human Review* as workload. This separation reflects deployment risk in V&V workflows, where false accepts can silently rewrite identifiers in Turtle artefacts and propagate semantic errors downstream.

At $\tau_{\mathrm{hr}} = 0.50$, AnchorMap attains AUTO-only precision 0.787, recall 0.842, and F1-score 0.814. The corresponding AUTO-only confusion matrix is TP $= 48$, false positive (FP) $= 13$, false negative (FN) $= 9$, and true negative (TN) $= 4$ ($\mathrm{N}_{\mathrm{AUTO}} = 74$), as shown in Figure 2. Out of the full dataset ($N = 144$), the system routes 61 as *Accept*, 70 as *Human Review*, and 13 as *No_Match*, corresponding to a human-review rate of 48.6%. Most

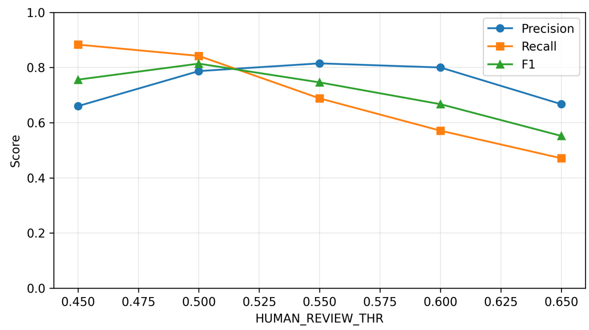

Figure 3: AUTO-Only Precision, Recall, and F1-Score vs. Human-Review Threshold τ_{hr}

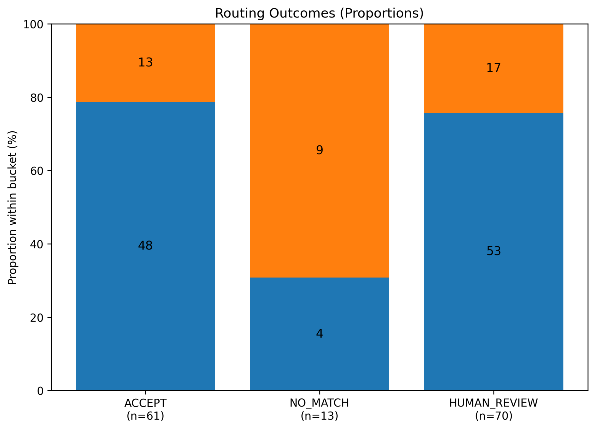

Figure 4: Routing Outcomes at $\tau_{\mathrm{hr}} = 0.50$ by Bucket (Accept, *No_Match*, Human Review) and Ground-Truth Correctness. Blue indicates ground-truth-aligned outcomes within each bucket; orange indicates the complementary outcomes.

remaining errors concentrate in the *Accept* bucket (false accepts), while *Human Review* captures the majority of ambiguous cases by design, and *No_Match* forms a small abstention set at this operating point.

The Variable Renaming Agent rewrites the 61 accepted variable instances across the 9 Turtle artefacts; Human Review items are deferred for adjudication and *No_Match* items are left unchanged by design.

Case studies

In addition to aggregate metrics, we inspect representative mapping instances at the selected operating point $\tau_{\mathrm{hr}} = 0.50$. Table 3 includes examples of correct automatic accepts, false positives among accepts, borderline cases routed to Human Review, and ground-truth negative cases that are correctly rejected as *No_Match*. These instances illustrate how routing decisions relate to confidence and candidate ambiguity, and they expose a recurring risk case where the reasoning step selects a plausible schema term even when the ground truth indicates that no canonical target exists.

Table 3: Representative Mapping Instances at $\tau_{\mathrm{hr}} = 0.50$ (A=Accept, HR=Human Review, NM=*No_Match*)

OEM label	Ground truth	System output	Route
`Output_kW`	`rated_output_power`	`rated_output_power`	A
`Speed_r_min`	`rotational_frequency`	`minimum_air_speed`	A
`stator_phase_inductance_Lpp`	`inductance`	`inductance`	HR
`temperature_adjustment_factor_fT`	`DONTEXIST`	`No Output`	NM

DISCUSSION

At the selected operating point ($\tau_{\mathrm{hr}} = 0.50$), Figures 3 and 4 show how τ_{hr} controls the balance between automation, human review workload, and deployment risk. The threshold τ_{hr} acts as a deployment risk-control mechanism: accepted mappings rewrite identifiers in downstream Turtle artefacts, so false positives can silently propagate into model configuration and analysis. In contrast, false negatives surface as explicit unmapped variables that are easier to detect during integration, which motivates prioritising precision over recall in this setting.

The sweep in Figure 3 highlights the automation–review trade-off: lowering τ_{hr} increases automation but admits more false accepts, whereas increasing τ_{hr} reduces commit risk at the cost of higher review workload. This behaviour is visible in Figure 4, where most incorrect outcomes occur among *Accept* decisions, while *Human Review* absorbs the majority of ambiguous cases by design. Across thresholds, a recurring failure mode is that the LLM selects a plausible schema term even when no valid canonical target exists. This behaviour is consistent with known hallucination tendencies in large language models (Huang et al., 2023; Lin et al., 2022) and motivates explicit abstention and conservative commit criteria.

As summarised in Table 1, AnchorMap differs from prior ontology and schema alignment studies by embedding matching within a practical engineering workflow. The target vocabulary is a versioned JSON-LD schema aligned with VIS that defines the permissible canonical identifiers, while selective automation is controlled via dual confidence thresholds and accepted mappings trigger executable changes in Turtle artefacts. This makes the cost of false positives materially higher than in offline alignment benchmarks and motivates robust abstention.

Future work should strengthen abstention and confidence calibration without assuming prior knowledge of ground-truth existence. As this study is intended as a proof of concept, future work can make abstention a first-class outcome in the LLM prompt, requiring *No_Match* selection when no candidate is justified. Acceptance criteria can also be tightened by requiring stronger separation margins or clearer agreement between the retrieval result and the candidate selected by the LLM. Confidence calibration methods and self-evaluation mechanisms (Ren et al., 2023; Wen et al., 2025) may further reduce false discovery risk. Future evaluation should use a larger and more diverse dataset, including additional subsystems, OEM sources, and document types, to better assess the generalisability of the approach beyond the electric motor case study. Ablations that compare retrieval-only and always-LLM variants could further isolate the value of conditional reasoning.

CONCLUSIONS

This paper presented AnchorMap, a schema-driven multi-agent pipeline for standardising differing engineering variable labels to a canonical vocabulary aligned with the Vessel Information Structure (VIS). The scientific contribution lies in combining embedding-based candidate retrieval with conditional LLM reasoning under a confidence-based routing policy with a human-in-the-loop mechanism. This architecture allows the system to commit only when confidence is sufficient, abstain when no justified match exists, and defer ambiguous cases to engineers' judgement, which makes the automation boundary transparent and adjustable through a parameter. The practical significance of this approach is that it enables variable-level interoperability without requiring organisations to change their internal naming conventions, and it reduces the manual effort needed to integrate different engineering models into co-simulation and collaborative design workflows. Overall, AnchorMap shows that selective automation grounded in a Single Source of Truth (SSOT) can produce more machine-verifiable variable identifiers for co-simulation and compliance workflows, while keeping semantic integrity risks explicitly controllable through thresholding and review. For reproducibility, the AnchorMap implementation and example artefacts are publicly available at `https://github.com/saadahmedrana/AnchorMap`.

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